

characteristics

- Transport eqn: $u_x + u_y = 0$; $u_t + cu_x = 0 \Rightarrow u(x,t) = f(x-ct) \Rightarrow$ translation in time $\rightarrow$ speed c .
- Constant coeff: $au_x + bu_y = 0 \Rightarrow \langle a, b \rangle \cdot \nabla u = 0$ slope = $\frac{b}{a} \Rightarrow y(x) = \frac{b}{a}x + C \Rightarrow u(x,y) = f(C) \Rightarrow u(x,y) = f(\frac{b}{a}x - by)$
- variable coeff: $y u_x + x u_y = 0 \Rightarrow \langle y, x \rangle \cdot \nabla u = 0$ slope = $\frac{dy}{dx} = \frac{x}{y} \Rightarrow \int y dy = \int x dx \Rightarrow y^2 = x^2 + C \Rightarrow u(x,y) = f(C) \Rightarrow u(x,y) = f(\frac{y^2 - x^2}{2})$
- coordinate method: e.g. $u_{xx} + u_{yy} = 1$
define $x' = x/y, y' = y-x \Rightarrow u_x = \frac{\partial u}{\partial x'} \frac{\partial x'}{\partial x} + \frac{\partial u}{\partial y'} \frac{\partial y'}{\partial x} = u_{x'} + u_{y'}$
 $\Rightarrow u_y = u_{x'} - u_{y'} \therefore$ PDE: $u_{x'} + u_{y'} + u_{x'} - u_{y'} = 1 \Rightarrow u_{x'} = \frac{1}{2} \Rightarrow u = \frac{1}{2}x' + C = \frac{1}{2}x' + f(y')$

→ well-posedness: soln. exists, is unique, and stable.
→ wave eqn: $u_{tt} = c^2 u_{xx} \Rightarrow (\partial_t - c \partial_x)(\partial_t + c \partial_x) u = 0 \Rightarrow \mathcal{L} = (\partial_{tt} - c^2 \partial_{xx})$

→ d'Alembert: $u_{tt} = c^2 u_{xx}$
cauchy { initial pos. $u(x,0) = \phi(x)$
initial v $u_t(x,0) = \psi(x)$ } $\Rightarrow u(x,t) = \frac{1}{2} [\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(s) ds$
→ waves have finite speed of propagation
dependence on (t_0, x_0)
influence (x_0, t_0)
Energy: $E(t) = \frac{1}{2} \int_{-\infty}^{\infty} (u_t)^2 + c^2 (u_x)^2 dx$; $E'(t) \leq 0$ (energy is non-increasing)
special function = u_t
 $u_{tt} = c^2 u_{xx} \Rightarrow \int_{-\infty}^{\infty} u_{tt} u_x dx = \int_{-\infty}^{\infty} c^2 u_{xx} u_t dx \Rightarrow \frac{d}{dt} \frac{1}{2} \int_{-\infty}^{\infty} (u_t)^2 dx = c^2 \left[\frac{d}{dt} \int_{-\infty}^{\infty} u_x u_t dx \right] = -\frac{d}{dt} \left(\frac{1}{2} \int_{-\infty}^{\infty} (u_x)^2 dx \right)$
 $\Rightarrow \frac{d}{dt} \left(\frac{1}{2} \int_{-\infty}^{\infty} (u_t)^2 dx \right) = c^2 \frac{d}{dt} \left(-\frac{1}{2} \int_{-\infty}^{\infty} (u_x)^2 dx \right) \Rightarrow \frac{d}{dt} \int_{-\infty}^{\infty} (u_t)^2 + c^2 (u_x)^2 dx = 0 \Rightarrow E' = 0$

→ uniqueness application: let $w = u - v \Rightarrow w_{tt} = c^2 w_{xx}$; $w(x,0) = \phi - \phi = 0$
 $w_t(x,0) = \psi - \psi = 0$
By energy methods $w=0 \Rightarrow u=v$ [$E(t) = E(0)$]

→ Heat/diffusion eqn: $u_t = k u_{xx} \Rightarrow \mathcal{L} = \partial_t - k \partial_{xx}$
is max. principle: maximum must be attained at $t=0$ or $x=0, x=l \rightarrow$ also min.
→ Energy: $E(t) = \frac{1}{2} \int_0^l [u(x,t)]^2 dx \rightarrow$ uniqueness: consider $w = u - v$
special function: $w_t = k w_{xx}$
 $w(x,0) = 0, w(0,t) = 0, w(l,t) = 0$
 $\Rightarrow \int_0^l w w_t dx = k \int_0^l w w_{xx} dx \Rightarrow \frac{d}{dt} \left(\frac{1}{2} \int_0^l w^2 dx \right) = -k \int_0^l (w_x)^2 dx \leq 0 \Rightarrow \int_0^l w^2 dx \leq \int_0^l w^2(0) dx = 0$
 $\therefore w = 0 \Rightarrow u = v$

→ Diffusion on whole line:
 $u_t = k u_{xx} \quad (-\infty < x < \infty, 0 < t < \infty)$
 $u(x,0) = \phi(x)$
 $\Rightarrow u(x,t) = \int_{-\infty}^{\infty} S(x-y,t) \phi(y) dy$; $S = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}}$
fundamental soln, Green function (Dirac Delta) at $t=0$
if $x \neq 0, \lim_{t \rightarrow 0} S(x,t) = 0$
 $\lim_{t \rightarrow 0} S(0,t) = \infty$
 $\int_{-\infty}^{\infty} S(x,t) dx = 1$
→ Diffusion on half line:
 $v_t - k v_{xx} = 0 \quad (0 < x < \infty, 0 < t < \infty)$
 $v(x,0) = \phi(x)$
 $v(0,t) = 0$
consider $\phi_{\text{odd}}(x) = \begin{cases} \phi(x) & x > 0 \\ -\phi(-x) & x < 0 \end{cases}$ and $\begin{cases} u_t - k u_{xx} = 0 \\ u(x,0) = \phi_{\text{odd}}(x) \end{cases}$
 $v(x,t) = u(x,t) \quad x > 0 \quad u(x,t) = \int_{-\infty}^{\infty} S(x-y,t) \phi_{\text{odd}}(y) dy$
 $= \int_0^{\infty} S(x-y,t) \phi(y) dy - \int_{-\infty}^0 S(x-y,t) \phi(-y) dy$
 $= \int_0^{\infty} S(x-y,t) \phi(y) dy - \int_0^{\infty} S(x+y,t) \phi(y) dy$
 $= \int_0^{\infty} [S(x-y,t) - S(x+y,t)] \phi(y) dy$ method of odd reflections

→ wave reflections
 $v_{tt} - c^2 v_{xx} = 0 \quad (0 < x < \infty, -\infty < t < \infty)$
 $v(x,0) = \phi(x), v_t(x,0) = \psi(x)$
 $v(0,t) = 0$
 $v(x,t) = u(x,t) = \frac{1}{2} [\phi_{\text{odd}}(x+ct) + \phi_{\text{odd}}(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(y) dy$
 $v(x,t) = \frac{1}{2} [\phi(ct+x) + \phi(ct-x)] + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(y) dy, \quad x > 0$

→ Diffusion with a source: $\begin{cases} u_t - u_{xx} = f(x, t) & (-\infty < x < \infty, t > 0) \\ u(x, 0) = \varphi(x) \end{cases}$ inhomogeneous

$$\Rightarrow u(x, t) = \int_{-\infty}^{\infty} S(x-y, t) \varphi(y) dy + \int_0^t \int_{-\infty}^{\infty} S(x-y, t-s) f(y, s) dy ds$$

→ Source on a half-line:

$$\begin{cases} u_t - u_{xx} = f(x, t) & (0 < x < \infty, t > 0) \\ u(0, t) = h(t) \\ u(x, 0) = \varphi(x) \end{cases} \quad \text{let } v(x, t) = u(x, t) - h(t) = \begin{cases} v_t - u_{xx} = f(x, t) - h'(t) \\ v(0, t) = 0 \\ v(x, 0) = \varphi(x) - h(0) \end{cases}$$

use method of reflections to solve

→ Waves with a source:

$$\begin{cases} u_t - c^2 u_{xx} = f(x, t) \\ u(x, 0) = \varphi(x) \\ u_t(x, 0) = \psi(x) \end{cases}$$

$$\Rightarrow u(x, t) = \frac{1}{2} [\varphi(x+ct) + \varphi(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(\eta) d\eta + \frac{1}{2c} \int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(\eta, s) dy ds$$

→ Source on a half line:

$$\begin{cases} v_t - c^2 v_{xx} = f(x, t) \\ v(x, 0) = \varphi(x) \\ v_t(x, 0) = \psi(x) \\ v(0, t) = h(t) \end{cases}$$

$$v(x, t) = \varphi \text{ term} + \psi \text{ term} + h(t - \frac{x}{c}) + \frac{1}{2c} \int_0^t \int_0^{x+c(t-s)} f(\eta, s) dy ds$$

→ Fourier transform:

$$\mathcal{F} f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-inx} f(x) dx = \hat{f}(n); \quad \mathcal{F}(u_x)(\xi) = i\xi(\mathcal{F}u(\xi))$$

$$\Rightarrow \mathcal{F}(u_t - u_{xx}) = \hat{u}_t - \hat{u}_{xx} = \hat{u}_t + \xi^2 \hat{u} = 0 \Rightarrow \hat{u}_t = -\xi^2 \hat{u} \rightarrow \text{ODE}$$

→ Boundary value (Dirichlet Problem) → Separation of variables

→ Waves:

$$\begin{cases} u_{tt} = c^2 u_{xx} \text{ for } 0 < x < l \\ u(0, t) = u(l, t) = 0 \\ u(x, 0) = \varphi(x); u_t(x, 0) = \psi(x) \end{cases} \quad \text{let } u(x, t) = X(x)T(t)$$

$$\Delta u = (u_{tt} - c^2 u_{xx}) = (X(x)T''(t) - c^2 X''(x)T(t)) = 0$$

$$X(0) = 0 \Rightarrow c=0 \Rightarrow X(x) = D \sin \beta x$$

$$X(l) = 0 \Rightarrow D \sin \beta l = 0 \Rightarrow \beta l = n\pi, n=1, 2, 3, \dots$$

$$\Rightarrow X_n(x) = D_n \sin \frac{n\pi x}{l}, n=1, 2, 3, \dots$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2 \text{ eigenvalues}$$

$$\Rightarrow u_n(x, t) = \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l}$$

$$\Rightarrow u(x, t) = \sum_{n=1}^{\infty} u_n(x, t), \quad \varphi(x) = \sum_n A_n \sin \frac{n\pi x}{l}$$

$$\psi(x) = \sum_n \frac{n\pi c}{l} B_n \sin \frac{n\pi x}{l}$$

→ Diffusion:

$$\begin{cases} u_t = u_{xx} & 0 < x < l \\ u(0, t) = u(l, t) = 0 \\ u(x, 0) = \varphi(x) \end{cases}$$

$$\text{let } u(x, t) = X(x)T(t) \Rightarrow \frac{T'}{T} = \frac{X''}{X} = -\lambda = -\beta^2$$

$$\Rightarrow u(x, t) = \sum_{n=1}^{\infty} A_n e^{-\left(\frac{n\pi}{l}\right)^2 t} \sin \frac{n\pi x}{l}, \quad \varphi(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l}$$

→ Fourier series

$$\text{coefficients: } A_n = \frac{1}{l} \int_{-l}^l \varphi(x) \cos \frac{n\pi x}{l} dx \quad (n=0, 1, 2, \dots)$$

$$A_n = \frac{1}{l} \int_{-l}^l \varphi(x) \sin \frac{n\pi x}{l} dx \quad (n=1, 2, 3, \dots)$$